

Sustainable Business Model Canvas

<u>Key Partners</u> • FAST NUCES • FESF • NGOs • Deaf Community Organizations • Technology Partners • Government and Educational Bodies <u>Something Else?</u> N/A.	<u>Key Activities</u> • Dataset preparation and annotation • Model training and optimization • Application development (mobile) • UI/UX design for accessibility • Testing and quality assurance <u>Key Resources</u> • Human resources (AI/ML developers, linguists, designers) • Technical resources (datasets, GPUs, cloud servers) • Software tools (TensorFlow, PyTorch, OpenCV, NLP libraries) • Partnerships (NGOs, Deaf community organizations)	<u>Value Propositions</u> • Real-time PSL translation bridging communication gaps • Interactive and engaging PSL learning platform • Affordable and accessible education for all users • Cultural preservation and awareness of PSL <u>High-Level Concept</u> • Duolingo for Sign Language	<u>Customer Relationships</u> • Personalized onboarding for new users • Progress tracking and reminders • Feedback-driven updates and improvements <u>5. Channels</u> • Mobile app stores (Google Play) • Social media outreach (Facebook, Instagram, Twitter) • Partnerships with schools, NGOs, and universities	<u>Customer Segments</u> • Deaf and hard-of-hearing individuals in Pakistan • Family, friends, and caregivers of the Deaf community • Educational institutions (schools, universities, training centers) • NGOs and organizations promoting inclusivity <u>Early Adopters</u> • Teachers and interpreters seeking digital tools • Parents of hearing-impaired children
<u>Cost Structure</u> • Infrastructure (Fixed): Hardware, dataset, licenses, R&D • People (Fixed): Consultation, employee training • Operations (Variable): Cloud storage, office & ops, product testing, maintenance			<u>Revenue Streams</u> • Institutional Licensing: NGOs, training centers	
<u>Eco-Social Costs</u> • Energy use from cloud servers (carbon footprint) • Accessibility gap if devices are unaffordable			<u>Eco-Social Benefits</u> • Promotes inclusivity for the Deaf community • Increases PSL awareness and education • Reduces communication barriers in society	